

Beyond the Threshold:

The Shift to Aseptic Predictive Intelligence

Aseptic Predictive Intelligence uses machine learning and Bayesian mathematics to classify and locate contamination sources in real-time — preventing batch rejections before a threshold is ever breached.

THE CHALLENGE

The "Silent Breach" Problem



Threshold Monitoring is Reactive

Traditional systems only trigger after limits are breached, leading to manual, "guesswork" investigations.



Data-Rich but Insight-Poor

Thousands of particle counts are logged every hour, yet their specific causes remain unidentified.



Contamination as a Signal

Contamination is not random — it is a *"Digital Fingerprint"* moving through a *fluid-dynamic grid*. Every particle has a name and a source.

Case Study — Invisible Bearing Failure (ISO 5 Filling Line)

85% Faster Response

Hybrid Forensic Model vs. Traditional SPC (Threshold-Based)

Traditional SPC	14.5 min — Batch at risk
Predictive Intelligence	2.2 min — Batch preserved

THE SOLUTION

The Dual-Engine Framework

⚙ Temporal Engine — WHAT HAPPENED?

Uses sktime to identify "Shapelet Signatures"

Unique discriminative patterns in time series data that identify the contamination mechanism — whether glove breach, mechanical friction, or HVAC drift — while the event is still unfolding.



Glove Breach

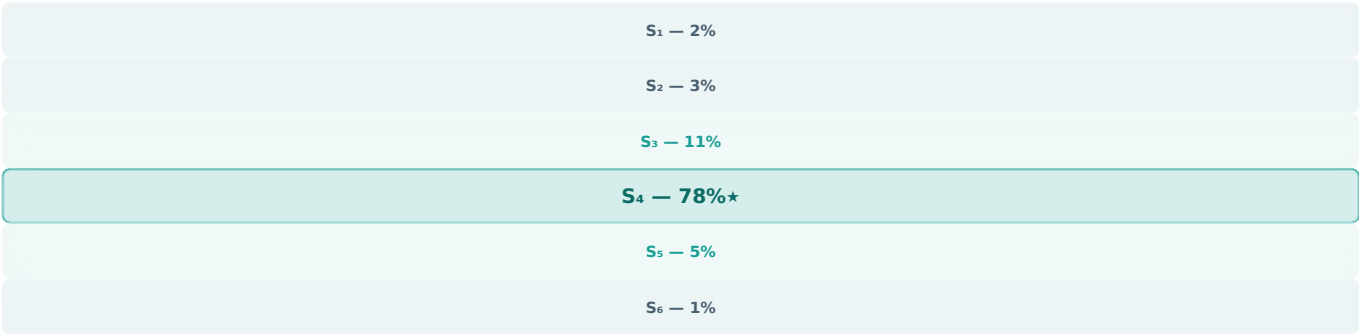
Sharp rise, slow decay



 Spatial Engine — WHERE IS IT?

Bayesian Search Theory — Source Triangulation

Sensor data and airflow dynamics are fed into a Bayesian probability grid. "Negative evidence" (clean upstream sensors) prunes impossible locations, narrowing a 3-metre filling line to a **20 cm target zone** — instantaneously.



★ Posterior probability converges at Sensor 4. Upstream sensors (S₁-S₃) provide negative evidence, eliminating upstream source hypotheses. No operator calculation required.

"Every particle has a name and a source. The data to find both is already being collected."

Paldron — Beyond the Threshold White Paper · Bayesian & sktime Framework for Predictive Contamination Control in Aseptic RABS